math

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October 2026

1 Introduction

Start from the chain rule:

$$\frac{\partial L}{\partial x_j} = \sum_i \frac{\partial L}{\partial s_i} \frac{\partial s_i}{\partial x_j}$$

Define

$$g_i = \frac{\partial L}{\partial s_i}$$

and plug in

$$\frac{\partial s_i}{\partial x_j} = s_i(\delta_{ij} - s_j)$$

so:

$$\frac{\partial L}{\partial x_j} = \sum_i g_i s_i (\delta_{ij} - s_j)$$

Expand:

$$= \sum_i g_i s_i \delta_{ij} - \sum_i g_i s_i s_j$$

Now, δ_{ij} is 1 only when $i = j$, so:

$$\sum_i g_i s_i \delta_{ij} = g_j s_j$$

And s_j does not depend on the summation index i, so:

$$\sum_i g_i s_i s_j = s_j \sum_i g_i s_i$$

Therefore:

$$\frac{\partial L}{\partial x_j} = g_j s_j - s_j \sum_i g_i s_i$$

Factor out s_j:

$$\boxed{\frac{\partial L}{\partial x_j} = s_j \left(g_j - \sum_i g_i s_i \right)}$$